Probability and Statistics

Hypothesis testing: Testing categorical associations

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Goal of this lecture

- Develop hypothesis tests and confidence intervals for categorical data (counts, proportions).
- Apply these tests to small datasets “by hand”.
- Learn to use R to apply the tests for larger data.

Overview

1. Hypothesis testing for proportions
2. Test for associations between two categorical variables
3. Tests for paired categorical data

Hypothesis testing for proportions

Case study

Saxony (Germany): 6155 new-born babies

- Boys: 3175 (51.6%)
- Girls: 2980 (48.4%)

Does this data indicate that the probability of having a boy is higher than 50%?





Dealing with count data

- Measurements are **counts** (3175 out of 6155): whole numbers.
- The proportion $\pi = 3175/6155 = 0.516$ is a number in $[0, 1]$.

Neither of these measures is normally distributed.

How can we express that under the null hypothesis we want the number of boys to be "not too far" from $6155/2 = 3077.5$?

Recall: binomial distribution

The number of times S an event occurs

- in a **fixed** number n of **independent** experiments
- where each time the event happens with probability p

follows a **binomial distribution** $B(n, p)$:

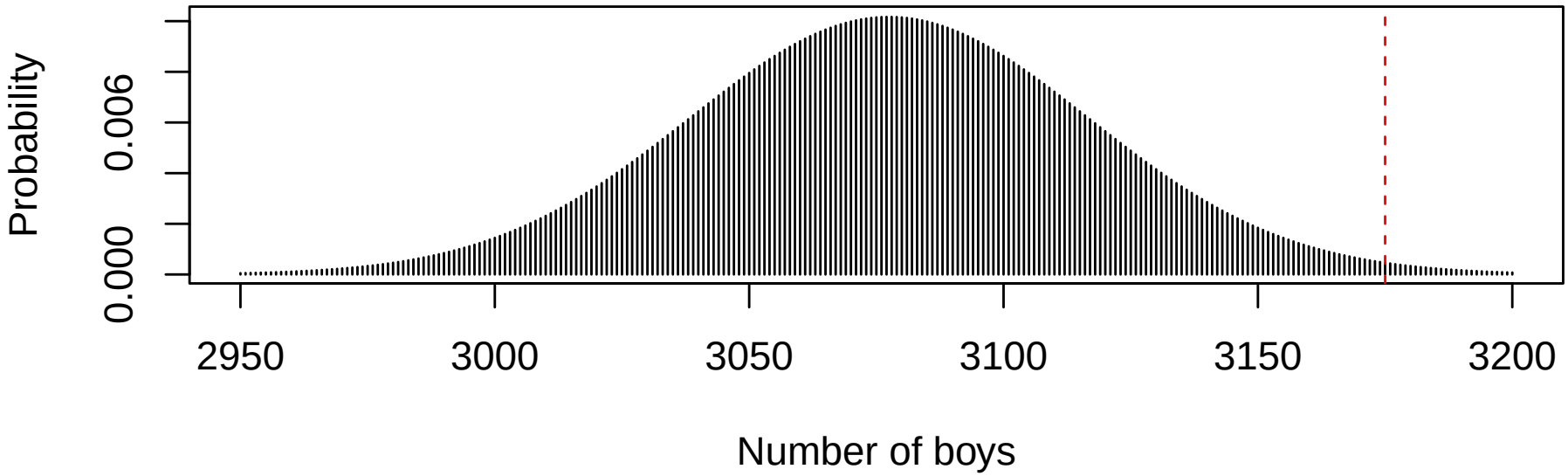
$$P(S = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

with mean $\mu = np$ and standard deviation $\sigma = \sqrt{np(1-p)}$.

Examples

- Number of students passing this course
- Number of boys out of 6155 births: $S \sim B(6155, p)$

What do we expect if boys and girls are equally likely?



- If $p = 1/2$, then $S \sim B(6155, 1/2)$, centered around 3077.5 (dashed line: observed 3175)
- Every single outcome is unlikely: $P(S = 3175) = 0.0005$, and even the most likely value has probability 0.01
- So we do not ask "how likely is 3175?", but "how likely is **3175 or more**?"

Elevated probability for boys? (Exact test)

- $H_0 : p = 0.5$ versus the **one-sided** alternative $H_A : p > 0.5$
- **Test statistic**: number of boys S, with

$$S \overset{H_0}{\sim} B(6155, 0.5)$$

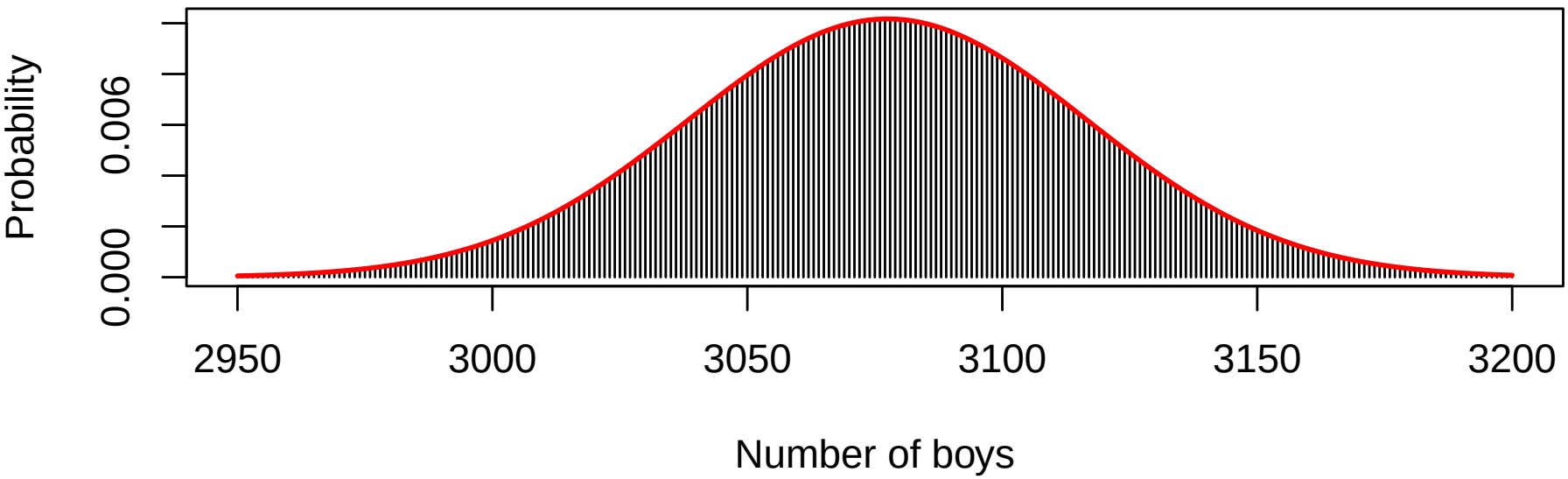
- p-value: probability to find, under H_0, a value at least as extreme as observed, in the direction of the alternative

$$\begin{aligned} P(S \geq 3175 \mid H_0) &= P(S = 3175) + ... + P(S = 6155) \\ &= \sum_{k=3175}^{6155} \binom{6155}{k} 0.5^{6155} \\ &= 0.00670 \end{aligned}$$

Elevated probability for boys? (Normal approximation)

Recall: $B(n, p)$ can be approximated by a normal distribution $N(\mu, \sigma^2)$, where

$$\mu = np = 3077.5 \quad \text{and} \quad \sigma = \sqrt{np(1-p)} = 39.2269.$$



For one-sided test

$$p = P(S \geq 3175) \approx P\left(Z \geq \frac{3175 - 3077.5}{39.2269}\right) = 0.0065.$$

Elevated probability for boys? (Two-sided)

- We then test null hypothesis $H_0 : p = 0.5$ **two-sided** with alternative $H_A : p \neq 0.5$
- Far from 3077.5 in **either** direction counts as extreme: 3175 is 97.5 above, so $6155 - 3175 = 2980$ is equally far below
- p-value is now twice as large:

$$p = P(S \geq 3175) + P(S \leq 2980) = 0.0134.$$

- Conclusion: evidence that the probability of a baby being a boy is different from 50%.

Binomial test in R

```r
binom.test(3175, 6155, p = 0.5)
```

```
    Exact binomial test

data:  3175 and 6155
number of successes = 3175, number of trials = 6155, p-
value = 0.0134
alternative hypothesis: true probability of success is not
95 percent confidence interval:
 0.5032696 0.5283969
sample estimates:
probability of success
             0.5158408
```

Use `alternative = "greater"` for the one-sided test.

One-sample test for proportions (binomial test)

Use of the one-sample test for proportions

Verify whether the proportion of experiments in which a certain event occurs deviates from a given value p_0

- Test $H_0 : p = p_0$ versus $H_A : p > p_0$ (one-sided) or $H_A : p \neq p_0$ (two-sided)
- Test statistic: number S of experiments where the event happens, with $S \overset{H_0}{\sim} B(n, p_0)$
- p-value for s observed events (one-sided):

$$P(S \geq s \mid p = p_0)$$

- Large n: use the normal approximation
- Condition: independent observations

95% CI for proportion

- **Proportion** $\pi = S/n$ of boys is approximately normally distributed, with

$$\mu = p \quad \text{and} \quad \sigma = \sqrt{\frac{p(1-p)}{n}}.$$

- We don't know p, but we can estimate it by π. This gives the 95% CI

$$\left[\pi - 1.96\sqrt{\frac{\pi(1-\pi)}{n}}, \pi + 1.96\sqrt{\frac{\pi(1-\pi)}{n}}\right]$$

- For the case study, $\pi = 0.516$ with 95% CI $[0.503, 0.528]$.
- 0.5 lies outside the CI, in agreement with the two-sided test ($p = 0.0134 < 0.05$)

Test for associations between two categorical variables

Genetic case-control study

Is mutation in BRCA1 gene associated with breast cancer?

Pro871Leu	Controls	Cases	Total
Pro/Pro	266	342	608
Pro/Leu	250	369	619
Leu/Leu	56	89	145
Total	572	800	1,372

- In **case-control studies** we choose a fixed number of cases and controls and investigate what they have been exposed to in the past
- Such studies are also called retrospective (see Chapter 2)

Same genotypes in cases and controls?

Percentage of each genotype within each group:

Pro871Leu	Controls	Cases	All subjects
Pro/Pro	46.5%	42.8%	44.3%
Pro/Leu	43.7%	46.1%	45.1%
Leu/Leu	9.8%	11.1%	10.6%
Total	100%	100%	100%

- H_0: no association, i.e. the genotype distribution is the same in cases and controls
- Under H_0, both groups should look like the "All subjects" column
- The percentages differ a little. Is this more than chance?

Expected counts under H_0

- Under H_0, 44.3% of the 572 controls should be Pro/Pro:

$$E_{11} = 572 \times \frac{608}{1,372} = 253.5$$

- In general, the expected count in cell (i, j) is

$$E_{ij} = \frac{\text{row total}_i \times \text{column total}_j}{n}$$

	Observed O_{ij}			Expected E_{ij}		
Pro871Leu	Cont.	Case	Total	Cont.	Case	Total
Pro/Pro	266	342	608	253.5	354.5	608
Pro/Leu	250	369	619	258.1	360.9	619
Leu/Leu	56	89	145	60.5	84.5	145
Total	572	800	1,372	572	800	1,372

Comparing observed and expected counts

For each cell, compute $(O_{ij} - E_{ij})^2/E_{ij}$:

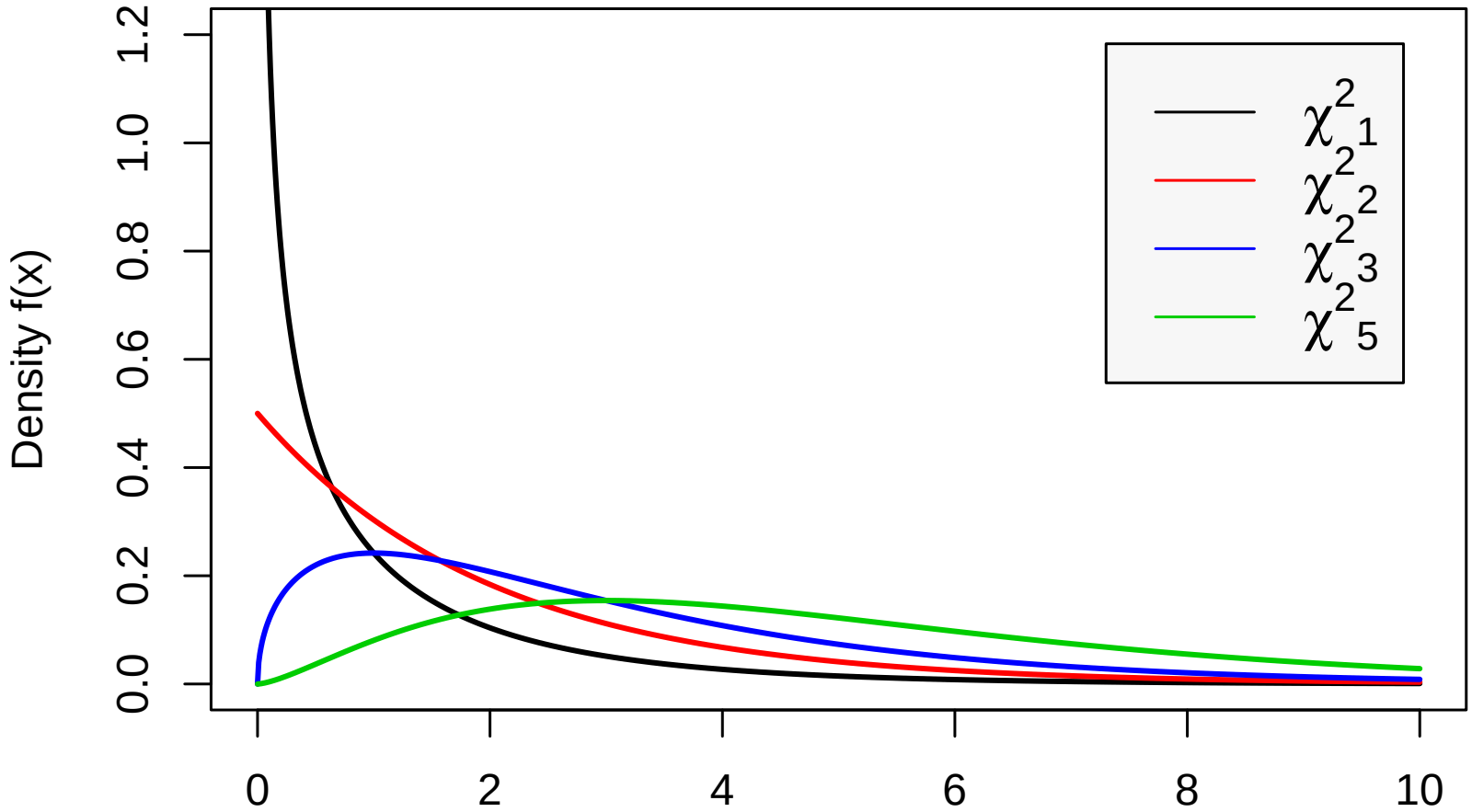
Pro871Leu	Controls	Cases
Pro/Pro	$(266 - 253.5)^2/253.5 = 0.62$	0.44
Pro/Leu	$(250 - 258.1)^2/258.1 = 0.25$	0.18
Leu/Leu	$(56 - 60.5)^2/60.5 = 0.33$	0.23

The test statistic is the sum over all cells:

$$X^2 = \sum_{i=1}^{3}\sum_{j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} = 2.055$$

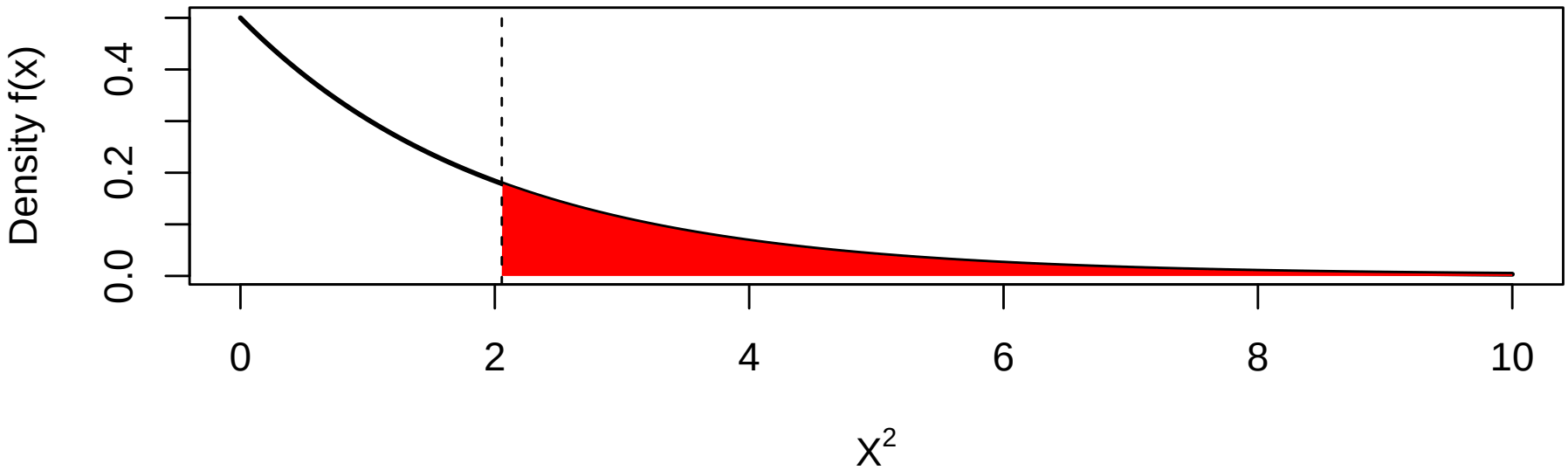
Large values of X^2 indicate deviation from H_0

X^2 follows a Chi-squared distribution



- In general: $(r-1) \times (c-1)$ degrees of freedom, with r the number of rows and c the number of columns
- Here: $(3-1) \times (2-1) = 2$

Computing the p-value



- Under H_0, X^2 follows a χ^2_2 distribution
- p-value: $P(X^2 \geq 2.055) = 0.358$ (red area)
- Only large values count against H_0: deviations in both directions are already included by squaring

Pearson's Chi-squared (χ^2) test

Use

Verify if 2 categorical variables X and Y are associated. Specifically, test if the distribution of a categorical variable is the same in 2 or more groups

- H_0 : X and Y independent, versus H_A : X and Y dependent
- Under H_0, test statistic

$$\sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}} \overset{H_0}{\sim} \chi^2_{(r-1)\times(c-1)}$$

follows Chi-squared distribution with $(r-1)\times(c-1)$ degrees of freedom, where r is number of rows and c is number of columns

Genetic case-control study: results

```r
m <- matrix(c(266,342,250,369,56,89), ncol=2, byrow=T)
chisq.test(m)
```

```
    Pearson's Chi-squared test

data:  m
X-squared = 2.0551, df = 2, p-value = 0.3579
```

- $p = 0.358 > 0.05$: no evidence that the Pro871Leu genotype is associated with breast cancer
- Consistent with the small differences in genotype percentages between cases and controls

Problem with case-control studies

Pro871Leu	Controls	Cases	Total
Pro/Pro	266	342	608
Pro/Leu	250	369	619
Leu/Leu	56	89	145
Total	572	800	1,372

- Impossible to determine probability for breast cancer
  - Naive data-analyst might say that estimated probability for breast cancer is 58% $\rightarrow$ **obviously not true!**
- Also relative risks, risk differences, $\cdots$ not possible
- Odds ratio is possible (see next slides)

Odds

- Odds for an event express how much more likely it is that the event happens than that it does not happen:

$$\frac{p}{1-p}$$

  with p the probability for the event

- Example: The odds for Leu/Leu among cases is

$$\text{odds}_1 = \frac{89/800}{711/800} = \frac{89}{711} = 0.125.$$

- The odds for Leu/Leu among controls is

$$\text{odds}_0 = \frac{56}{516} = 0.109.$$

Odds Ratio

- **Odds ratio (OR)** compares odds for an event between 2 groups:

$$OR = \frac{\text{odds for event when exposed}}{\text{odds for event when not exposed}}.$$

- When probability for event small in both groups ($< 5\%$), then

$$\text{odds} = p/(1-p) \approx p$$

and thus

$$OR \approx RR = \frac{\text{probability for event if exposed}}{\text{probability for event if not exposed}}.$$

- OR is always further away from 1 than relative risk (RR)

Odds ratio for the case study

- Odds ratio for Leu/Leu, cases versus controls:

$$OR = \frac{\text{odds}_1}{\text{odds}_0} = \frac{89/711}{56/516} = 1.15$$

- The odds ratio is symmetric: we get the same value for the odds of breast cancer, Leu/Leu versus other genotypes:

$$\frac{89/56}{711/516} = \frac{89 \times 516}{56 \times 711} = 1.15$$

- This is why the odds ratio can be estimated in a case-control study
- Interpretation: odds for breast cancer are 15% higher with mutation (Leu/Leu) than without
- A small effect, in line with the non-significant Chi-squared test ($p = 0.358$)

Yates' correction and Fisher's exact test

- Pearson's Chi-squared test assumes
  - independent observations
  - "large" sample: expected counts should be larger than 5 in at least 80% of cells
- If sample too small:
  - **Pearson's Chi-squared test with Yates' continuity correction** for 2×2 tables

$$\sum_{ij} \frac{(|O_{ij} - E_{ij}| - 1/2)^2}{E_{ij}} \overset{H_0}{\sim} \chi^2_1$$

  - **Fisher's exact test**

Tests for paired categorical data

Matched case-control study

Is a history of spontaneous abortion (SA) associated with heart disease?

- 50 women with heart disease (**cases**)
- Each case is **matched** with a similar woman without heart disease (**control**), e.g. of the same age
- For every woman we record: history of SA, yes or no?

! Paired data

The data consists of **50 pairs**, not of 100 independent women. Within a pair, case and control are similar by design.

Recall: for continuous data, we used the **paired t-test** for this kind of design.

Four kinds of pairs

Case	Control	Number of pairs	
SA	SA	7	concordant
SA	no SA	**18**	**discordant**
no SA	SA	**5**	**discordant**
no SA	no SA	20	concordant
Total		50	

- **Concordant** pairs: case and control have the same SA history
- **Discordant** pairs: case and control have a different SA history

The same data as a table of pairs

Case ↓ Control →	SA	no SA	Total
SA	$e = 7$	$\mathbf{f = 18}$	25
no SA	$\mathbf{g = 5}$	$h = 20$	25
Total	12	38	50

- Each cell counts **pairs**, not women
- Row totals: 25 out of 50 cases have a history of SA
- Column totals: 12 out of 50 controls have a history of SA

Why not Pearson's Chi-squared test?

It is tempting to count women instead of pairs:

	SA	no SA	Total
Cases	25	25	50
Controls	12	38	50

- Pearson's Chi-squared test assumes **independent** observations
- Here, each case is linked to a control: the observations are **dependent**

$\Rightarrow$ We need a test that works with the **pairs**.

Only discordant pairs carry information

Difference in the proportion with SA between cases and controls:

$$\hat{\pi}_{\text{cases}} - \hat{\pi}_{\text{controls}} = \frac{e+f}{n} - \frac{e+g}{n} = \frac{f-g}{n} = \frac{18-5}{50} = 0.26$$

- The concordant pairs (e and h) drop out
- A pair where both women had SA (or neither did) tells us nothing about the **difference** between cases and controls
- All the information is in f versus g: **18 against 5**

The key idea: a coin flip

Under H_0: SA is not associated with heart disease.

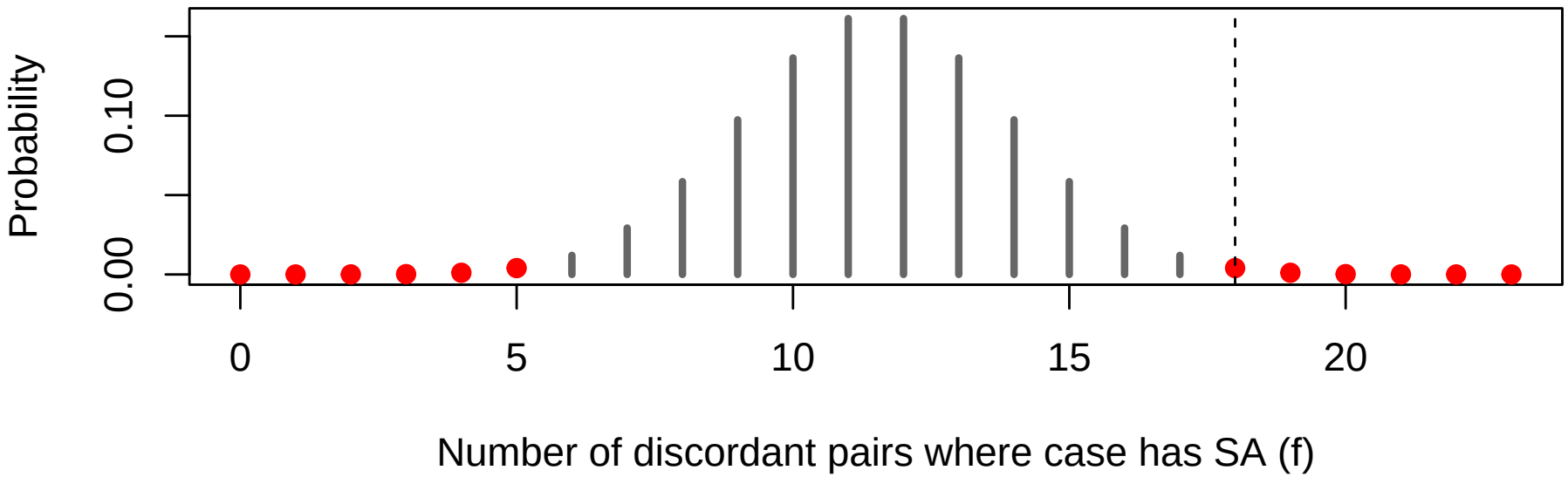
- In a discordant pair, exactly one of the two women has a history of SA
- Under H_0, it is equally likely to be the case or the control
- So among the $f + g = 23$ discordant pairs, the number f where the case has SA is

$$f \overset{H_0}{\sim} B(f + g, 1/2) = B(23, 1/2)$$

McNemar's test in one sentence

McNemar's test is a **binomial test** on the discordant pairs, with $H_0 : p = 1/2$.

Exact test



- Observed: $f = 18$ out of 23
- One-sided p-value: $P(f \geq 18) = 0.0053$
- Two-sided p-value: $2 \times P(f \geq 18) = 0.0106$ (red: $f \geq 18$ or $f \leq 5$)

Normal approximation: McNemar's test statistic

As for the Saxony babies, approximate $B(f+g, 1/2)$ by a normal distribution with

$$\mu = \frac{f+g}{2} \quad \text{and} \quad \sigma = \frac{\sqrt{f+g}}{2}.$$

Standardizing f gives **McNemar's test statistic**:

$$z = \frac{f-(f+g)/2}{\sqrt{f+g}/2} = \frac{f-g}{\sqrt{f+g}} \qquad \text{(continuity correction: } z = \frac{|f-g|-1}{\sqrt{f+g}}\text{)}$$

- Case study: $z = (13-1)/\sqrt{23} = 2.50$ and $p = 2 \times P(Z > 2.50) = 0.0123$
- Approximation is good if $f \times g/(f+g) \geq 5$
- Here $18 \times 5/23 = 3.9 < 5$: prefer the **exact test**

Conclusion for the case study

- Two-sided p-value (exact test): $0.0106 < 0.05$
- Heart disease is associated with a history of spontaneous abortion
- Size of the effect:
  - In 18 out of 23 discordant pairs (78%), it was the case who had a history of SA
  - Proportion with SA: 50% among cases versus 24% among controls

McNemar's test

Use

Verify if the probability of an event is the same in 2 **paired** groups

- H_0: the probability of the event is the same in both groups
- Exact test: $f \overset{H_0}{\sim} B(f+g, 1/2)$
- Large samples: $\frac{|f-g|-1}{\sqrt{f+g}} \overset{H_0}{\sim} N(0,1)$, if $f \times g/(f+g) \geq 5$
- Conditions: independent pairs
- Other paired designs: the same subject before and after treatment, twins, the color of tiger beetles in spring and in summer, $\cdots$

McNemar's test in R

```r
m <- matrix(c(7, 18, 5, 20), nrow = 2, byrow = TRUE)
mcnemar.test(m)
```

```
	McNemar's Chi-squared test with continuity correction

data:  m
McNemar's chi-squared = 6.2609, df = 1, p-value = 0.01234
```

```r
binom.test(18, 23)$p.value
```

```
[1] 0.01062202
```

Note: *R reports the square of our continuity-corrected test statistic, compared with a* χ^2_1 *distribution. The two-sided p-value is identical.*

Overview tests for categorical outcome

- Compare proportion with number: **binomial test**
- Compare proportions between 2 groups:
  - paired groups: **McNemar's test** (exact binomial test on the discordant pairs)
  - unpaired groups: **Pearson's Chi-squared test** (Fisher's exact test)
- Compare categorical measurement between at least 2 groups: **Pearson's Chi-squared test** (Fisher's exact test)

Summary

- **Binomial test**: compare a proportion with a given value; use the normal approximation for large n
- Confidence interval for a proportion: $\pi \pm 1.96\sqrt{\pi(1-\pi)/n}$
- In case-control studies, risks cannot be estimated, but **odds ratios** can
- **Pearson's Chi-squared test**: association between two categorical variables, comparing observed and expected counts
  - Small samples: Yates' correction or **Fisher's exact test**
- **McNemar's test**: paired categorical data; only the **discordant pairs** carry information, and under H_0 each one is a fair coin flip